

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 2 (Generated practice paper - Set 12)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

**0 1 . 1**

Explain **four** properties of water that make it essential for life.

**[4 marks]**

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**0 1 . 2**

Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

**[4 marks]**

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**0 1 . 3**

Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

**[2 marks]**

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0 2 . 1

Explain what a carbon footprint should include when calculated for a car.

[3 marks]

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0 2 . 2

Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

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0 2 . 3

Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

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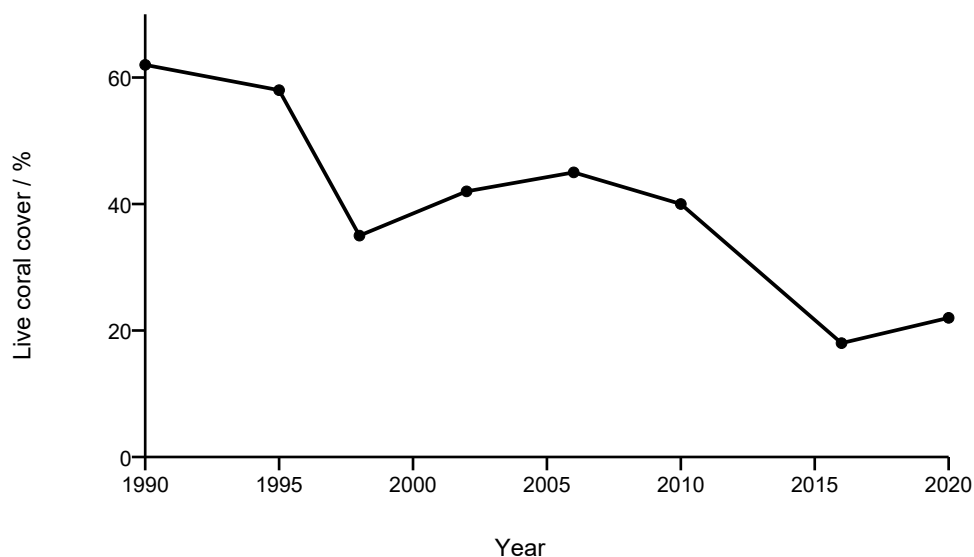
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03

Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

**Figure 1**



03

1

Describe the changes in coral cover shown in Figure 1.

[2 marks]

03

2

Explain what coral bleaching is and give **three** causes.

[4 marks]

03 . 3

Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

03 . 4

Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

10

04 . 1

Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

04 . 2

Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

1 .....

2 .....

3 .....

4 .....

0 4 . 3

Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

1 .....

2 .....

3 .....

10

0 5

Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 5 . 1

Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

| Species and adaptation                               | Design application  |  |
|------------------------------------------------------|---------------------|--|
| Tubercles (bumps) on the flippers of humpback whales |                     |  |
|                                                      | Self-cleaning glass |  |
| Shark skin scales that reduce friction               |                     |  |
|                                                      | Velcro              |  |

0 5 . 2

Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

0 5 . 3

Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

1

2

0 5 . 4

Explain why bird wing bones are a useful model for engineers.

[1 mark]

10

**0 6** . **1**

Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

**[1 mark]**

A Environmental DNA (eDNA) analysis

☐

B Radio tracking

☐

C Lincoln index (mark-release-recapture)

☐

D Leg rings

☐

**0 6** . **2**

Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

**[2 marks]**

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**0 6** . **3**

Explain how DNA databases could help to control the illegal trade in elephant ivory.

**[2 marks]**

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**5**

**0 7** . **1**

Explain the difference between hard and soft release and state the types of species for which each is suitable.

**[3 marks]**

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**0 7 . 2** Describe **five** features of a suitable release site.

**[5 marks]**

**1** .....

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**0 7 . 3** Explain the importance of post-release monitoring.

**[2 marks]**

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**10**

**0 8** Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

**Table 1**

| Feature                                                         | Mixed indigenous woodland   | Sitka spruce plantation      |
|-----------------------------------------------------------------|-----------------------------|------------------------------|
| Tree species                                                    | 12                          | 1                            |
| Age structure                                                   | All ages, dead wood present | All planted in the same year |
| Planting density                                                | Natural, variable           | Close, 2 m spacing           |
| Timber yield / m <sup>3</sup> ha <sup>-1</sup> yr <sup>-1</sup> | 4                           | 16                           |
| Bird species                                                    | 38                          | 9                            |
| Ground flora species                                            | 60                          | 4                            |

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| 0 | 8 | . | 1 |
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Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

**[6 marks]**

|   |   |   |   |
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Explain why commercial plantations are felled before the trees reach full size.

**[2 marks]**

|   |   |   |   |
|---|---|---|---|
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Explain why there is little demand today for coppiced or pollarded wood.

**[2 marks]**

**10**

**0 9**

Table 1 shows the energy inputs and food energy outputs of three farming systems.

**Table 1**

| <b>System</b>                                               | <b>Energy input / GJ ha<sup>-1</sup> yr<sup>-1</sup></b> | <b>Food energy output / GJ ha<sup>-1</sup> yr<sup>-1</sup></b> |
|-------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| Extensive cereal farming (low inputs)                       | 4                                                        | 20                                                             |
| Intensive cereal farming (fertiliser, pesticide, machinery) | 20                                                       | 60                                                             |
| Intensive beef (grain-fed)                                  | 40                                                       | 4                                                              |

**0 9****. 1**

Calculate the energy ratio (output / input) for each system.

**[2 marks]****Answer**

**0 9****. 2**

Use your answers to explain the difference between productivity and efficiency.

**[3 marks]**

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**0 9****. 3**Explain what is meant by an **energy subsidy** and give **three** examples.**[3 marks]**

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09 . 4

Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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10 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

[illegible]

**[1 mark]**

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**Write an essay on one of the following topics.**

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

Discuss how human activities influence biodiversity, using examples from a range of different contexts.

**[25 marks]**

**OR**

**1 1 . 2**

Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.





[illegible]

**END OF QUESTIONS**